

SMART HOSTEL AND ROOM ALLOCATION SYSTEM

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YEAR-2ND

SEMESTER-4

SESSION-SPRING 2026

OOPs PROJECT REPORT

PROBLEM STATEMENT

Manual hostel room allocation systems are widely used in many institutions, but they often suffer from inefficiencies and lack of accuracy. These systems typically rely on manual record keeping, which increases the chances of errors such as duplicate entries, incorrect room assignments, and overbooking of rooms. Additionally, manual systems lack transparency and are difficult to maintain, especially when the number of students increases.

The objective of this project is to design and implement a Smart Hostel & Room Allocation System using Object-Oriented Programming principles. The system aims to automate student registration and room allocation while ensuring that constraints such as room capacity are strictly followed. By introducing a structured and modular design, the system improves efficiency, reduces human errors, and provides a scalable solution that can be extended in the future.

SYSTEM DESIGN

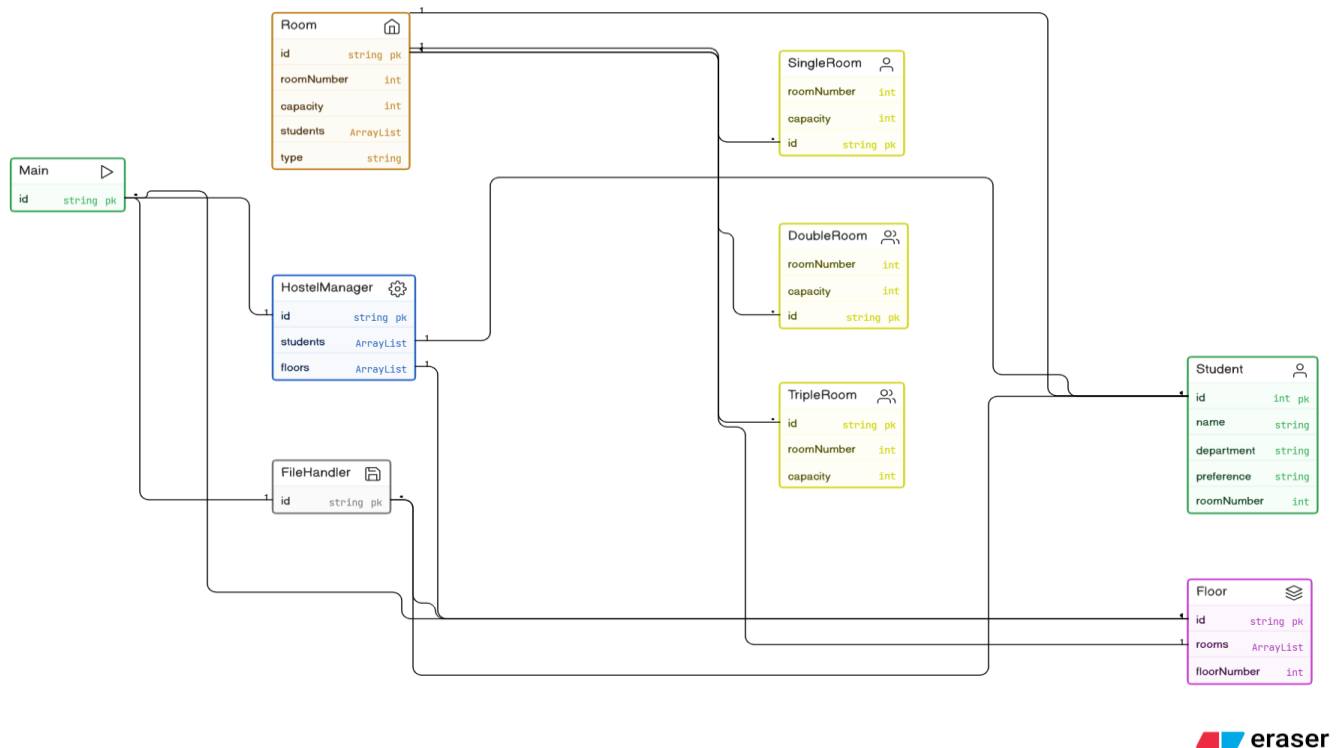
The system is designed using a modular and object-oriented approach, where responsibilities are clearly divided among different classes. The architecture follows a layered design consisting of model and controller components. The model layer includes classes such as Student and Room, which represent real-world entities, while the controller layer manages user interaction and system flow.

The major classes in the system include Student, Room, and HostelManagementSystem. The Student class stores student-related information such as ID and name. The Room class manages room details including room number, capacity, and the list of students assigned to it. It also contains important methods such as addStudent() and isFull() to handle allocation logic. The HostelManagementSystem class acts as the controller, handling user input, displaying menus, and coordinating between different objects.

The package structure is organized logically to improve readability and maintainability. Classes are separated into different files such as Student.java, Room.java, and Main.java. This separation

ensures that each class has a single responsibility and makes the system easier to extend.

The interaction flow begins when the user starts the program and selects an option from the menu. Based on the input, the system performs operations such as adding a student or allocating a room. During allocation, the system checks room capacity and assigns the student accordingly. Finally, the updated information is displayed to the user.



OOP CONCEPTS DEMONSTRATED

Encapsulation is implemented in the Student and Room classes, where data members are declared as private and accessed through public methods. This ensures that the internal state of objects cannot be modified directly, thereby improving data security and integrity.

Abstraction is applied in the room allocation process. The complexity of checking room capacity and managing student lists is hidden inside methods such as `addStudent()` and `isFull()`. The user interacts with the system without needing to understand the internal logic, making the system easier to use.

Inheritance is demonstrated through the design of room types such as `SingleRoom`, `DoubleRoom`, and `TripleRoom`, which can extend the base `Room` class. This allows common properties to be reused while enabling specific behaviors for different room types.

Polymorphism allows methods to behave differently depending on the object type. For example, allocation logic can vary based on room type, providing flexibility in system behavior.

The system follows a modular design pattern with clear separation of concerns. The controller manages system flow, while model classes handle data and logic. This improves maintainability and scalability.

IMPLEMENTATION HIGHLIGHTS

- SMART ROOM ALLOCATION –

```
27
28     for (Floor f : floors) {
29         for (Room r : f.rooms) {
30
31             if (s.getPreference().equalsIgnoreCase("Single") && r instanceof SingleRoom && !r.isFull())
32                 preferredRooms.add(r);
33
34             else if (s.getPreference().equalsIgnoreCase("Double") && r instanceof DoubleRoom && !r.isFull())
35                 preferredRooms.add(r);
36
37             else if (s.getPreference().equalsIgnoreCase("Triple") && r instanceof TripleRoom && !r.isFull())
38                 preferredRooms.add(r);
39         }
40     }
```

This demonstrates intelligent allocation logic using user preference, type checking (`instanceof`), and capacity constraints, ensuring efficient and realistic room assignment.

- FILE HANDLING –

```
5
6  ✓ static void saveStudents(ArrayList<Student> students) {
7  ✓     try {
8           BufferedWriter bw = new BufferedWriter(new FileWriter("students.txt"));
9
10  ✓         for (Student s : students) {
11               bw.write(
12                   s.getId() + "," +
13                   s.getName() + "," +
14                   s.getDepartment() + "," +
15                   s.getPreference() + "," +
16                   s.getRoomNumber()
17               );
18               bw.newLine();
19         }
20     }
```

This ensures data persistence, allowing the system to store student and allocation data in a file, making the system usable across multiple sessions.

- UNIQUE ID VALIDATION –

```
13     boolean isDuplicateId(int id) {
14         for (Student s : students) {
15             if (s.getId() == id) {
16                 return true;
17             }
18         }
19         return false;
20     }
21 }
```

This enforces data integrity by ensuring each student has a unique ID, preventing duplication and maintaining consistency in the system.

- ROOM CHANGE –

```
139
140  ✓ if (newRoom == null) {
141       System.out.println("New room not found.");
142       return;
143   }
144
145  ✓ if (newRoom.isFull()) {
146       System.out.println("New room is full.");
147       return;
148   }
149
150   oldRoom.removeStudent(found);
151   newRoom.addStudent(found);
152
153   System.out.println("Room changed successfully.");
154 }
155
```

This demonstrates dynamic state management, where a student is safely removed from one room and added to another, reflecting real-world room transfer operations.

- EXCEPTION HANDLING –

```
73
74         try {
75             choice = sc.nextInt();
76         } catch (Exception e) {
77             System.out.println("Invalid input!");
78             sc.next();
79             continue;
80         }
81
```

Prevents runtime crashes and ensures robust user input handling.

TESTING & ERROR HANDLING

Test cases considered

Normal input case: Valid student details are entered. The system successfully adds the student and allocates a room.

Multiple student allocation: Multiple students with different preferences are added. The system allocates rooms based on preference and availability.

Preference-based allocation: When a valid preference (Single/Double/Triple) is given, the system assigns a matching room if available.

Random allocation case: If the preference is invalid or unavailable, the system assigns a random available room.

Search functionality: When a valid student ID is entered, the correct student details are displayed.

Remove (vacate) functionality: When a valid student ID is entered, the student is removed and the room occupancy is updated.

Room change functionality: When a valid student ID and room number are given, the student is moved to the new room if space is available.

Role-based access: Admin has full access, while user has restricted access. Unauthorized actions are blocked.

File handling: After restarting the program, student data and room allocations are successfully restored.

Edge cases handled

Duplicate student ID: If the same ID is entered again, the system prevents duplication and asks for a new ID.

Invalid name input: If the name contains numbers or symbols, the input is rejected and the user is prompted again.

Invalid department selection: If an invalid option is chosen, the system asks the user to re-enter a valid choice.

Invalid preference input: If the preference is not Single, Double, or Triple, the system assigns a random room.

Room full condition: If all rooms are occupied, the system displays that no rooms are available.

Room change to full room: If the target room is full, the system does not allow the change.

Non-existent student search/remove: If the ID does not exist, the system displays “Student not found”.

Invalid menu input: If a non-integer or out-of-range value is entered, the system handles it and prompts again.

Failure Scenarios

Invalid input type: If a string is entered instead of a number, exception handling prevents the program from crashing and asks for input again.

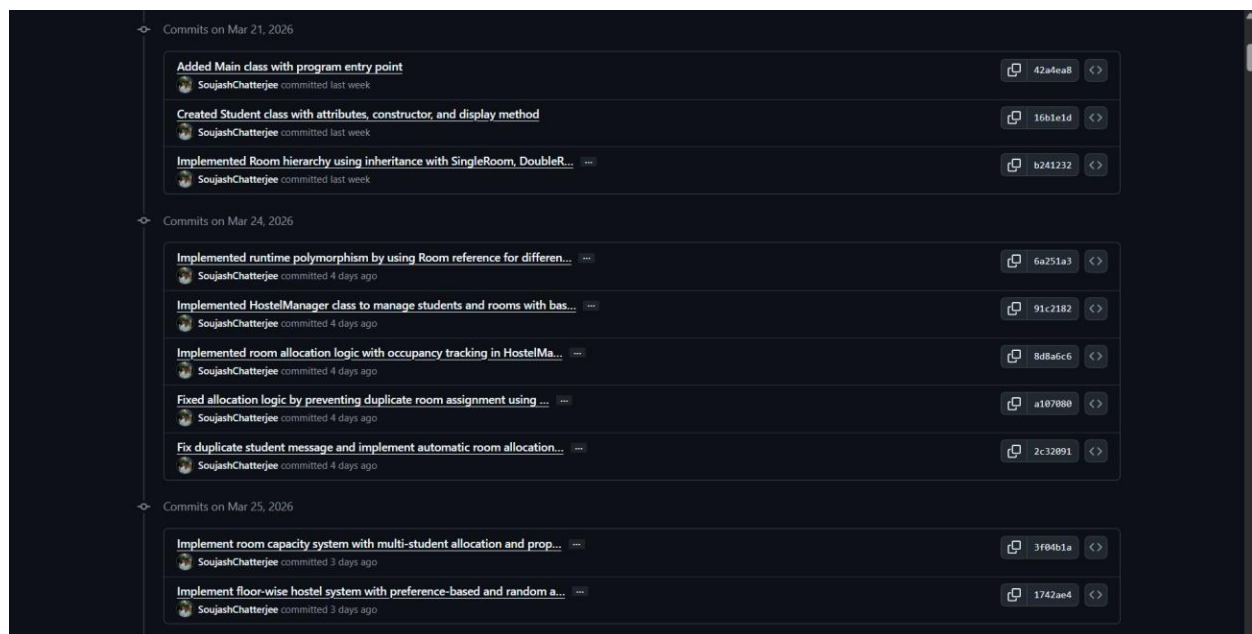
File not found (first run): If the data file does not exist, the system skips loading without crashing.

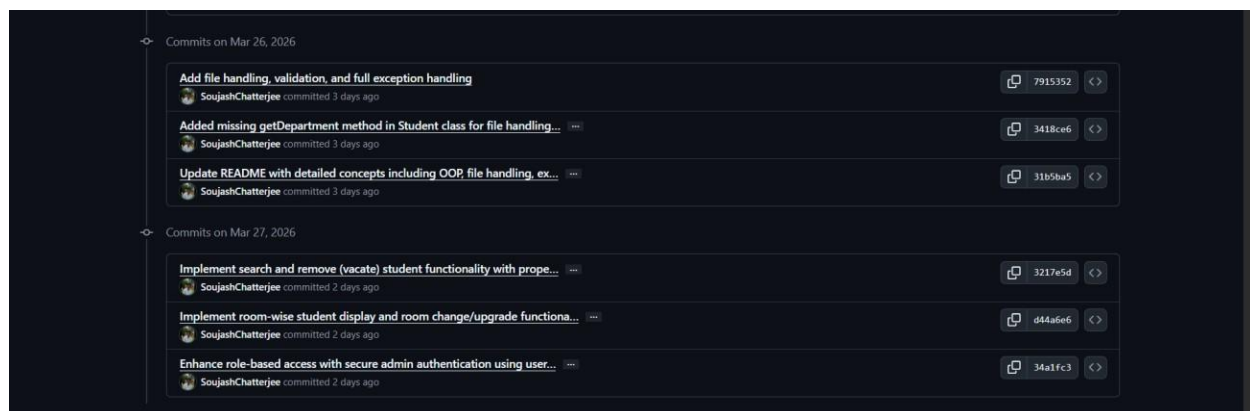
File read/write errors: If there is an I/O issue, an error message is shown and the program continues.

Unauthorized access attempt: If a user tries to access admin features, the system denies access.

Incorrect login credentials: If wrong admin credentials are entered, the system logs in as a normal user.

GIT WORKFLOW





The project was developed in a step-by-step manner, starting with the creation of the main entry point using the Main class to establish the basic execution flow and menu-driven interface. After that, the Student class was created to represent student data with attributes like name and ID along with display functionality.

Next, the room structure was implemented using object-oriented principles. A base Room class was created and extended into SingleRoom, DoubleRoom, and TripleRoom, demonstrating inheritance and polymorphism. Runtime polymorphism was further applied by handling different room types through a common reference.

The HostelManager class was then introduced to manage all core operations such as adding students, allocating rooms, and maintaining records. Initially, a basic room allocation system was implemented, which was later improved by adding occupancy tracking and fixing issues like duplicate assignments.

The system was then enhanced by introducing room capacity logic, allowing multiple students in a room based on type. A floor-wise structure was also added, organizing rooms under different floors to make the system more realistic and scalable.

Further improvements included preference-based allocation, where students could choose room types, along with a fallback mechanism that assigns rooms randomly if the preferred option is not available. This added intelligent decision-making to the system.

Validation and exception handling were implemented to ensure robustness. This included checking for duplicate student IDs, validating names and inputs, and handling incorrect data entries without crashing the program.

File handling was added to enable data persistence. Student details along with room allocations were saved to a file and reloaded when the program restarted, ensuring continuity of data.

Additional features such as searching for a student, removing (vacating) a student, displaying room-wise student details, and changing rooms were implemented to make the system more functional and closer to real-world applications.

Finally, role-based access control was introduced with a secure admin login using username and password. This restricted critical operations to authorized users only, adding a basic level of security.

Overall, the project evolved from a simple structure into a complete system with object-oriented design, intelligent allocation logic, persistent storage, validation, and security features.

CONCLUSION & FUTURE SCOPE

The Smart Hostel & Room Allocation System successfully demonstrates the application of Object-Oriented Programming concepts to solve a real-world problem. The system provides an efficient and reliable way to manage student accommodation and room allocation.

The modular design and use of OOP principles make the system easy to maintain and extend. Future enhancements can include the addition of a graphical user interface, integration with a database, and implementation of advanced authentication mechanisms. Overall, the project serves as a strong example of how structured programming and design principles can be used to build practical and scalable software solutions.

APPENDIX

INITIAL PROJECT PROPOSAL :

PROJECT PROPOSAL

Name: Soujash Chatterjee

Department: CSE IOTCSBT

Roll Number: 48

Project Title: Smart Hostel and Room Allocation System

Problem Statement

Managing hostel room allocation manually in colleges often leads to inefficiencies and errors in record keeping. Administrators find it difficult to track room availability, assign rooms to students, and update hostel records. The Smart Hostel and Room Allocation System aims to automate this process by maintaining student information, room availability, and allocation records in an organized manner. The system allows administrators to add students, allocate rooms, view occupancy status, and update records when students vacate rooms. The project applies object-oriented programming concepts to build a modular and maintainable hostel management solution.

Target Users

1. Hostel wardens
2. Hostel administrators
3. College management responsible for hostel operations

Core Features

1. Add and manage student information
2. Allocate rooms based on availability
3. View hostel room availability and occupancy status
4. Update or remove student records when required
5. Display list of students and room details
6. Input validation and exception handling for incorrect entries

7. Store and manage records using collections

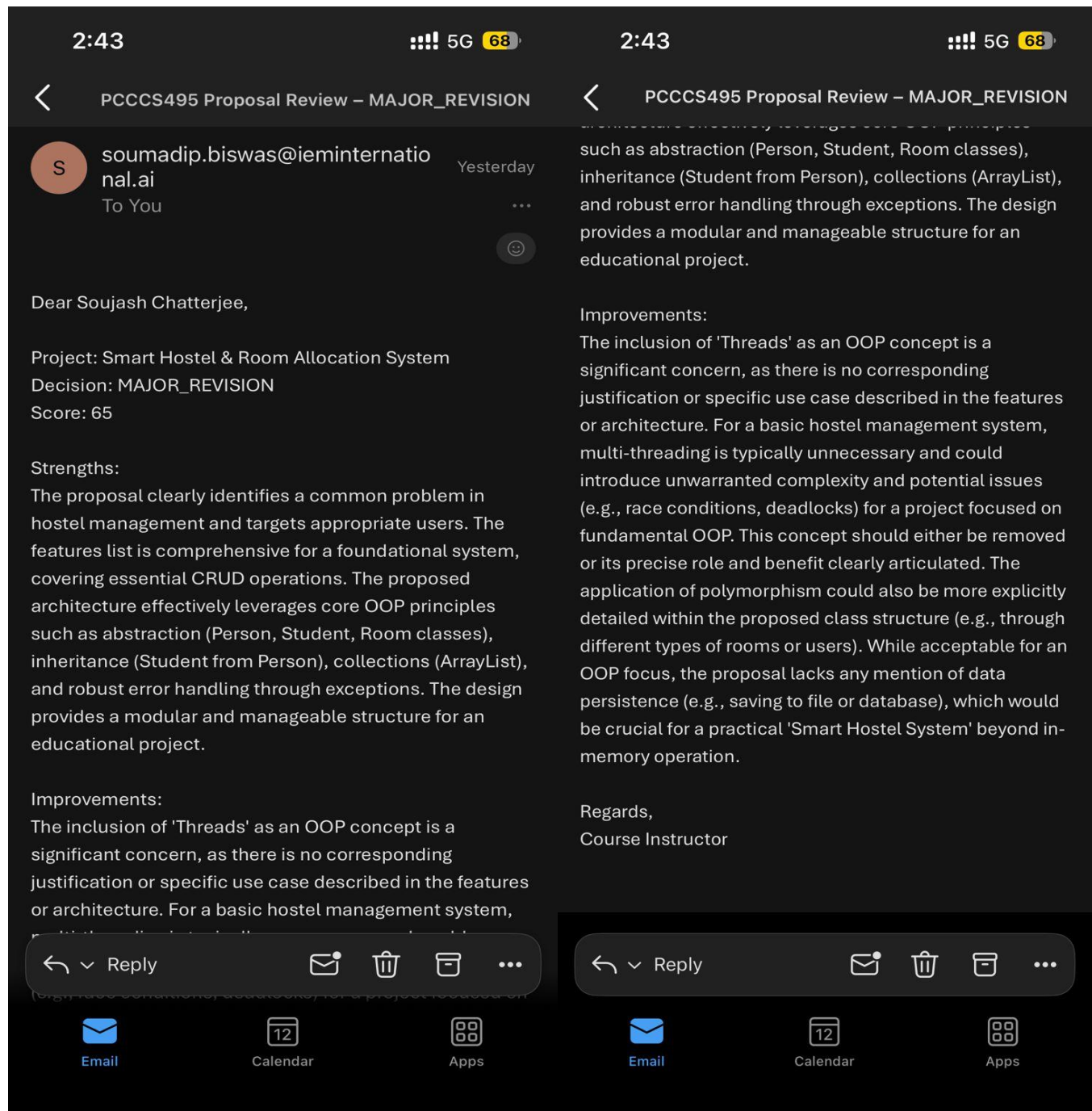
OOP Concepts to be Used

1. Abstraction: A base abstract class such as Person will define common attributes like name and ID.
2. Inheritance: Classes such as Student and Warden will inherit from the Person class.
3. Polymorphism: Methods such as displayInfo() will behave differently depending on the object type.
4. Exception Handling: Used to manage invalid input and incorrect room allocation data.
5. Collections: Student and room records will be stored using ArrayList.

Proposed Architecture Description

The system will consist of multiple classes such as Person, Student, Room, HostelManager, and Main. Person will act as the base class containing common attributes. Student will inherit from Person and store student-specific details. The Room class will maintain room number and occupancy status. HostelManager will control the core operations such as adding students, allocating rooms, and displaying records. Collections will store data and exception handling will ensure stable program execution. This modular architecture improves maintainability and scalability.

REVIEW MAIL RECEIVED :



SUBSEQUENT SUBMISSION :

Project Proposal

Name- Soujash Chatterjee , CSE – IoTCSBT , Roll no -48

Project Title- Smart Hostel and Room Allocation System

1. Problem Statement

Hostel room allocation in many colleges is still managed manually using registers or spreadsheets. This manual process often leads to errors in record keeping, difficulty in tracking room availability, and inefficiencies in updating student information. Administrators frequently face challenges such as identifying vacant rooms, updating student records when rooms are vacated, and retrieving information quickly when required. The Smart Hostel and Room Allocation System aims to automate hostel management by maintaining student information, room availability, and allocation records in a structured and organized way. The system will enable administrators to easily manage student entries, allocate rooms based on availability, update records, and track hostel occupancy using an object-oriented software design.

2. Target Users

- Hostel Wardens
- Hostel Administrators
- College Management responsible for hostel operations

3. Core Features

- Add and manage student information
- Allocate hostel rooms based on availability
- Display available and occupied rooms
- Update room status when students vacate rooms
- Remove or update student records
- Display a list of students with their allocated rooms
- Input validation and exception handling to prevent invalid entries
- Store hostel data persistently using file storage

4. OOP Concepts to be Used

Abstraction: An abstract base class Person will define common attributes such as name and ID.

Inheritance: Classes such as Student and Warden will inherit properties from the Person class.

Polymorphism: Different room types such as SingleRoom, DoubleRoom, and TripleRoom will override methods such as displayRoomDetails() to demonstrate runtime polymorphism..

Encapsulation: Student and room details will be stored as private attributes and accessed using getter and setter methods.

Exception Handling: Exception handling will be used to manage invalid inputs and incorrect room allocation attempts.

Collections: Java ArrayList will be used to store student and room records dynamically.

File Handling : The system will use Java File Handling or Serialization to save student and room data to files so that records remain available even after the program restarts.

5. Proposed System Architecture

Person (Abstract Class)

- Attributes: name, ID
- Method: displayInfo()

Student Class

- Inherits from Person
- Stores student-specific information such as department and allocated room

Warden Class

- Inherits from Person
- Represents hostel administrators managing room allocation

Room Class

- Attributes: room number, occupancy status
- Maintains information about assigned student

HostelManager Class

Handles core system operations such as:

- Adding students
- Allocating rooms
- Updating room status
- Displaying hostel records
- Saving and loading data

Main Class

- Provides a menu-driven interface for interacting with the system